

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q24-2440

Blanchard 138 kV
July 29, 2025

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q24-2440. The Interconnection Customer proposes a 78.6 MW solar plus storage facility interconnecting at the Blanchard 138 kV substation in the New Jersey area. The request was received on August 12, 2024. The study evaluated system intact and single contingency conditions using the regional planning case, including 477 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$2,291,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Hudson – White Rock-Nike 138kV	Crawfordsville Power Plant – Pine Tree Acres 138kV	120.2	326	6.1
Suntree – W Caldwell 138kV line	W Orange – Magnolia 1708 138kV	103.9	298	10.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Hudson – White Rock-Nike 138kV	Rebuild line with higher-capacity conductor	\$1,663,000	100% (\$1,663,000)
Suntree – W Caldwell 138kV line	Reconductor line	\$2,245,000	28% (\$628,000)
	Total allocated to project		\$2,291,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.